ATAL TINKERING LAB COMPONENT LIST

Year: 2026

01-MICROCONTROLLERS-AND-DEV-BOARDS

Item #	Component / Board Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**ESP32 DevKit V1 (30-Pin / 38-Pin)**	32-bit Dual-Core Xtensa LX6 @ 240 MHz; onboard 2.4 GHz Wi-Fi + Bluetooth BLE 4.2/5.0; 520 KB SRAM; 4 MB Flash; 3.3V logic; capacitive touch, ADC, DAC, PWM on all GPIOs	**16 Units**	**Consumable**
02	**Seeed Studio XIAO ESP32-S3 Development Board**	WiFi & Bluetooth 5.0 Microcontroller Board with integrated battery charging; ESP32-S3 dual-core Xtensa 32-bit LX7 @ 240 MHz; ultra-compact thumb-sized form factor; ideal for IoT, robotics, wearable & smart device projects	**5 Units**	**Consumable**
03	**Espressif ESP32-S3-DevKitC-1-N8R8 Development Board**	Flagship ESP32-S3 development board featuring ESP32-S3-WROOM-1 module; 8MB Octal SPI Flash + 8MB Octal PSRAM; 2.4 GHz Wi-Fi (802.11 b/g/n) + Bluetooth 5 (LE); dual USB-C ports (OTG + UART); hardware vector instructions for TinyML/Edge AI	**5 Units**	**Consumable**
04	**Arduino Uno R3 (or equivalent)**	ATmega328P 8-bit AVR; 5V; 14 Digital I/O (6 PWM); 6 Analog inputs; 16 MHz; includes protective casing and USB Type-A to Type-B programming cable	**30 Units**	**Consumable**
05	**Arduino Nano (or equivalent)**	ATmega328P 8-bit AVR; 5V; breadboard DIP form factor; 14 Digital I/O; 8 Analog inputs; Mini-USB cable included	**5 Units**	**Consumable**
06	**Arduino Mega 2560 (or equivalent)**	ATmega2560 8-bit AVR; 5V; 54 Digital I/O (15 PWM); 16 Analog inputs; 4 UART serial ports; USB cable included	**5 Units**	**Consumable**
07	**Raspberry Pi 3 Model B+**	Quad Core 64-bit Broadcom ARM @ 1.4 GHz; 1 GB RAM; Wi-Fi & BLE; 40-pin GPIO; 4x USB; HDMI; 5V 2.5A power supply & protective enclosure	**4 Units**	**Consumable**
08	**Raspberry Pi 5 (2GB RAM)**	64-bit Quad-Core ARM Cortex-A76 @ 2.4 GHz; 2 GB LPDDR4X SDRAM; VideoCore VII GPU; Dual 4Kp60 micro-HDMI; PCIe 2.0 interface; Gigabit Ethernet, Dual-band 802.11ac Wi-Fi, Bluetooth 5.0 / BLE; 2x USB 3.0, 2x USB 2.0; USB-C 5V 5A PD input	**2 Units**	**Consumable**
09	**Raspberry Pi 4 Model B (2GB RAM)**	64-bit Quad-Core ARM Cortex-A72 @ 1.5 GHz; 2 GB LPDDR4 SDRAM; Dual 4Kp60 micro-HDMI displays; Gigabit Ethernet; 2.4/5.0 GHz Wi-Fi, Bluetooth 5.0; 2x USB 3.0, 2x USB 2.0; USB-C 5V 3A input	**2 Units**	**Consumable**
10	**Raspberry Pi Zero (v1.3)**	Single-Core ARM11 @ 1 GHz (BCM2835); 512 MB LPDDR2 RAM; mini-HDMI video; micro-USB OTG data port; micro-USB power; CSI camera connector; ultra-compact 65×30mm board size	**2 Units**	**Consumable**
11	**Raspberry Pi Zero W (with WiFi)**	Single-Core ARM11 @ 1 GHz; 512 MB RAM; onboard 802.11 b/g/n Wi-Fi + Bluetooth 4.1 / BLE; mini-HDMI; micro-USB OTG port; CSI camera connector; ultra-compact form factor	**2 Units**	**Consumable**
12	**Raspberry Pi Zero 2 W**	64-bit Quad-Core ARM Cortex-A53 @ 1 GHz (RP3A0 SiP); 512 MB LPDDR2 RAM; onboard 2.4 GHz Wi-Fi + Bluetooth 4.2 / BLE; mini-HDMI; micro-USB OTG; CSI-2 camera connector; up to 5x faster performance	**2 Units**	**Consumable**
13	**HIKVISION Extreme 16GB microSDXC Memory Card (Class 10, V30)**	High-speed 16GB Class 10, V30 MicroSD card; 92 MB/s read speed, 15 MB/s write speed; compatible with smartphones, cameras, CCTV; drop protection; 3-year warranty; **Dedicated for Raspberry Pi 3B+ Single-Board Computer (Operating System Boot & Local Storage)**	**5 Units**	**Consumable**
14	**NodeMCU v3 (ESP8266)**	ESP8266 Wi-Fi SoC with onboard CP2102/CH340 USB-UART; 3.3V logic; integrated TCP/IP stack; PCB trace antenna	**10 Units**	**Consumable**
15	**WeMos D1 R2 (ESP8266)**	Arduino Uno form-factor Wi-Fi development board powered by ESP8266; 5V–12V barrel jack input; 11 Digital I/O; 1 Analog input	**5 Units**	**Consumable**

02-BASIC-ELECTRONIC-COMPONENTS-AND-ICS

Item #	Discrete Component Name	Technical Specifications & Model Details	Practical Lab / Circuit Role	Required Quantity	Asset Classification
01	**Bare 5V Sugar-Cube Electromagnetic Relay**	Songle SRD-05VDC-SL-C; 5-pin SPDT bare relay; contacts rated 10A 250V AC / 30V DC *(Requires manual transistor & diode driver)*	Learning coil actuation, transistor switching & flyback diode protection	**5 Units**	**Consumable**
02	**Single 1-Digit 7-Segment LED Display (Common Cathode)**	0.56-inch 1-digit red 7-segment display; 10-pin DIP through-hole; forward voltage 2.0V *(Requires 7-8 current-limiting resistors)*	Learning direct binary-to-decimal LED segment decoding	**5 Units**	**Consumable**
03	**Single 1-Digit 7-Segment LED Display (Common Anode)**	0.56-inch 1-digit red 7-segment display; 10-pin DIP through-hole; active-low configuration	Comparative study of active-high vs active-low segment logic	**5 Units**	**Consumable**
04	**Bare 4-Digit 7-Segment LED Display**	12-pin bare multiplexed 4-digit display *(No TM1637 driver IC; requires multiplexed pin scanning)*	Teaching display multiplexing, ghosting prevention & persistence of vision	**5 Units**	**Consumable**
05	**Bare 16×2 Character LCD Panel (HD44780)**	16-pin parallel interface LCD display; 5V DC; blue backlight *(Raw display without I2C backpack board)*	Teaching parallel data buses (4-bit/8-bit), RS, Enable & RW pin control	**5 Units**	**Consumable**
06	**10-Segment LED Bargraph Display Array**	20-pin DIP array with 10 individually addressable red/green LED segments	Analog level indicators, battery level meters & audio VU meters	**5 Units**	**Consumable**
07	**Bare LDR Photoresistors (GL5528)**	5mm cadmium sulfide (CdS) light-dependent resistor; dark resistance 1 MΩ, light resistance 10-20 kΩ	Voltage divider circuits for automatic light/dark detectors	**5 Units**	**Consumable**
08	**Bare NTC Thermistors (10kΩ Bead)**	2-pin negative temperature coefficient thermistor; B(25/50) = 3950 K; resistance decreases as heat rises	Building analog thermal detection and fire alarm circuits	**5 Units**	**Consumable**
09	**Rotary Carbon Potentiometers (10kΩ)**	3-pin panel mount / breadboard rotary potentiometer with knurled shaft and matching control knobs	Dial controls for PWM motor speed, tone pitch & LCD contrast	**5 Units**	**Consumable**
10	**Breadboard Miniature Trimpots (10kΩ - 3362P)**	3-pin single-turn horizontal/vertical cermet trimmer potentiometer	Precision threshold calibration on prototype breadboards	**5 Units**	**Consumable**
11	**ATmega328P-PU Standalone IC (with 16MHz Crystal Kit)**	28-pin DIP AVR microcontroller bare chip paired with 16 MHz HC-49S quartz crystal and two 22pF ceramic load capacitors	Building a "DIY Arduino on Breadboard / Perfboard" from scratch	**5 Sets**	**Consumable**
12	**ATmega16U2 Standalone IC**	8-bit AVR microcontroller bare IC in DIP/SOIC package; 16 KB flash; USB controller capability; 2.7V - 5.5V	Advanced USB microcontroller prototyping & bare IC design	**3 Units**	**Consumable**
13	**Shift Register IC (74HC595)**	16-pin DIP 8-bit serial-in parallel-out shift register with 3-state output latches	Controlling 8 to 64 LEDs or multi-digit displays using only 3 GPIO pins	**5 Units**	**Consumable**
14	**Dual Operational Amplifier IC (LM358)**	8-pin DIP dual low-power general-purpose op-amp; single supply 3V - 32V	Building analog signal amplifiers, active filters & sensor preamps	**5 Units**	**Consumable**
15	**Dual Voltage Comparator IC (LM393)**	8-pin DIP dual precision voltage comparator; open-collector output	Understanding analog voltage threshold slicing (how sensor modules work)	**5 Units**	**Consumable**
16	**Timer IC — LM555 / NE555**	Precision timing bipolar IC; DIP-8 package; operating voltage 4.5V - 16V	Astable multivibrators (clock pulse generators) and monostable timers	**5 Units**	**Consumable**
17	**Linear Voltage Regulators (LM78XX)**	3-terminal linear regulators in TO-220 package: LM7805 (+5V), LM7809 (+9V), LM7812 (+12V); 1.5A peak	Designing clean regulated DC power supply rails on breadboards	**5 Units**	**Consumable**

Item #	Discrete Component Name	Technical Specifications & Model Details	Practical Lab / Circuit Role	Required Quantity	Asset Classification
18	**Logic-Level Power N-Channel MOSFET (IRLZ44N / IRF540N)**	TO-220 power MOSFET; 55V, 47A, low R(DS(on)); logic-level gate threshold (V(GS(th)) = 1.0V - 2.0V)	Direct high-speed switching of high-current DC motors, coils & 12V LED strips	**5 Units**	**Consumable**
19	**Bare Dual H-Bridge Motor Driver IC (L293D)**	16-pin DIP bare driver IC (unmounted without shield); 600mA per channel; internal clamp diodes	Breadboard motor control circuits and H-bridge theory	**5 Units**	**Consumable**
20	**Bare Darlington Transistor Array IC (ULN2003A)**	16-pin DIP array of 7 high-voltage, high-current NPN Darlington pairs with common suppression diodes	Driving unmounted stepper motors, relays, and solenoid coils	**5 Units**	**Consumable**
21	**Assorted Resistor Kit (1/4W Carbon Film)**	30 resistors each across 20 standard values: 10 Ω, 100 Ω, 220 Ω, 330 Ω, 470 Ω, 1 kΩ, 4.7 kΩ, 10 kΩ, 100 kΩ, 1 MΩ, etc.	Current-limiting for LEDs, pull-up/pull-down, and voltage dividers	**5 Kits (3,000 pcs)**	**Consumable**
22	**Assorted Capacitor Kit (Ceramic & Electrolytic)**	Assorted values: 22 pF, 100 pF, 10 nF, 100 nF (104), 1 µF, 10 µF, 100 µF, 1000 µF; 16V–50V	Noise filtering, crystal loading, decoupling & timing circuits	**5 Kits (1,000 pcs)**	**Consumable**
23	**Diodes and Transistors Kit**	BC547 / 2N3904 (NPN), BC557 / 2N3906 (PNP), 1N4148 (Fast signal), 1N4001 / 1N4007 (1A Rectifier)	Amplification, logic inverter switching, rectification & flyback protection	**3 Kits (75 pcs)**	**Consumable**
24	**Zener Diodes Assortment (3.3V, 5.1V, 12V)**	0.5W axial leaded Zener voltage reference diodes	Voltage clamping, regulation & basic 5V-to-3.3V logic level shifting	**3 Packs (30 pcs)**	**Consumable**
25	**Through-Hole Diffused LEDs (5mm) — Red**	Standard 5mm round diffused red LED; forward voltage 1.8V - 2.2V; current 20 mA	Visual output indicators and logic state signals	**5 Packs (250 pcs)**	**Consumable**
26	**Through-Hole Diffused LEDs (5mm) — Green**	Standard 5mm round diffused green LED; forward voltage 2.0V - 2.4V; current 20 mA	Power/Go status indicators	**5 Packs (250 pcs)**	**Consumable**
27	**Through-Hole Diffused LEDs (5mm) — Blue**	Standard 5mm round diffused blue LED; forward voltage 2.8V - 3.2V; current 20 mA	Special indicator and primary RGB color mixing	**5 Packs (250 pcs)**	**Consumable**
28	**Through-Hole Diffused LEDs (5mm) — Yellow**	Standard 5mm round diffused yellow LED; forward voltage 2.0V - 2.4V; current 20 mA	Warning and caution state signaling	**5 Packs (250 pcs)**	**Consumable**
29	**Through-Hole Clear LEDs (5mm) — White**	Ultra-bright 5mm clear white LED; forward voltage 3.0V - 3.4V; current 20 mA	Illumination and miniature flashlight projects	**5 Packs (250 pcs)**	**Consumable**
30	**RGB LEDs (5mm Common Cathode, 4-Pin)**	4-pin through-hole 5mm diffused RGB LED; common ground pin *(Requires 3 individual resistors)*	Color blending, PWM color mixing & status multi-color alerts	**3 Packs (60 pcs)**	**Consumable**
31	**Tactile Momentary Push Buttons (6×6mm)**	4-pin breadboard-compatible momentary micro push buttons with colored caps	Manual digital inputs, reset triggers & interface buttons	**3 Packs (150 pcs)**	**Consumable**
32	**Miniature SPDT Slide Switches**	3-pin through-hole sub-miniature slide switches; ON-ON 2-position rating 0.5A 50V DC	Power master switching and hardware mode selection	**2 Packs (40 pcs)**	**Consumable**
33	**DIP Switch Array (4-Position & 8-Position)**	Standard 2.54mm pitch breadboard rocker/slide DIP switch blocks	Binary address setting, mode configuration & logic truth table tests	**5 Packs (20 pcs)**	**Consumable**
34	**Bare Tilt Ball Switch Sensor (SW-520D)**	2-pin brass-cased rolling ball tilt sensor switch	Simple gravity/tilt orientation detection without software libraries	**5 Units**	**Consumable**
35	**Bare Electret Microphone Capsule (9×7mm)**	2-pin raw electret microphone cartridge with solder pads *(Requires pull-up & coupling capacitor)*	Building custom acoustic sensor & sound trigger circuits	**5 Units**	**Consumable**
36	**Piezoelectric Transducer Disc Plate**	40mm copper-backed piezoelectric ceramic disc element with pre-soldered positive/negative wire leads	Vibration energy harvesting, knock detection & contact microphones	**5 Units**	**Consumable**

Item #	Discrete Component Name	Technical Specifications & Model Details	Practical Lab / Circuit Role	Required Quantity	Asset Classification
37	**Precision Analog Temperature Sensor (LM35)**	Precision linear analog temperature sensor; TO-92 3-pin transistor package; 10 mV/°C	Raw analog voltage temperature reading	**5 Units**	**Consumable**
38	**Force Sensitive Resistor (FSR)**	1.75× 1.5 inch square polymer thick-film pressure sensor; resistance varies with mechanical force	Pressure sensing, electronic scales & footstep pads	**3 Units**	**Consumable**
39	**Flexion Bend Sensor**	2.2-inch flexible carbon-resistive bend sensor; nominal flat resistance 25 kΩ	Robotic finger joint tracking and wearable glove controls	**3 Units**	**Consumable**
40	**Miniature Active Buzzer — Small**	5V DC active piezoelectric buzzer with internal oscillator (2-pin through-hole)	Audible alarms and beep notifications	**5 Units**	**Consumable**
41	**Loud Active Buzzer — Big**	5V–12V DC high-decibel active buzzer (2-lead round casing, 85+ dB output)	Perimeter intrusion alarms and fire warning buzzers	**5 Units**	**Consumable**
42	**Miniature Haptic Vibrating Motor**	10g coin/cylindrical vibration motor (1.5V - 3.7V DC) *(Requires transistor & flyback diode)*	Silent tactile alerts and wearable haptic feedback	**5 Units**	**Consumable**
43	**IR Optical Sensor Pair (TSOP1738 + 940nm LED)**	38 kHz infrared receiver IC (3-pin) paired with 940nm high-output IR emitting LED (2-pin)	Remote control decoders, IR beam breaks & optical tachometers	**5 Pairs**	**Consumable**

03-SENSOR-MODULES

Item #	Sensor Module Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**Digital Temperature & Humidity (DHT11)**	Single-bus digital output sensor module with pull-up resistor onboard; temp range 0–50°C; humidity range 20–90% RH; supply 3.3V–5V	**10 Units**	**Consumable**
02	**Analog Humidity Sensor Module**	Resistance-based relative humidity sensor probe board; range 20–95% RH; linear analog voltage response	**10 Units**	**Consumable**
03	**Soil Moisture Sensor Module**	Double-sided nickel-plated fork probe with LM393 comparator breakout module; analog and digital outputs	**8 Units**	**Consumable**
04	**Raindrop Detection Board**	50×40 mm nickel-plated rain sensor board with separate control module (LM393 comparator and trimmer)	**8 Units**	**Consumable**
05	**Turbine Water Flow Sensor**	Hall-effect liquid flow meter; 1–30 L/min range; 5V DC; pulse frequency output (f = 7.5× Q)	**5 Units**	**Consumable**
06	**PIR Human Motion Detector (HC-SR501)**	Pyroelectric infrared sensor module; dome Fresnel lens; range 3–7m; adjustable delay and sensitivity; 3.3V logic output	**6 Units**	**Consumable**
07	**Ultrasonic Distance Sensor (HC-SR04)**	40 kHz ultrasonic transmitter and receiver module; range 2 cm – 400 cm; precision 3mm; 4-pin header (VCC, Trig, Echo, GND)	**12 Units**	**Consumable**
08	**IR Proximity / Obstacle Avoidance Module**	940nm IR emitter and photodiode pair; onboard LM393 comparator; detection distance 2–30 cm; 35° angle	**8 Units**	**Consumable**
09	**8-Channel IR Tracking Sensor Array**	8-sensor reflective infrared bar with onboard comparators; 8 individual digital output pins for line-following robotics	**10 Units**	**Consumable**
10	**Sound Detection Microphone Module**	Electret condenser microphone with LM393 amplifier and digital threshold sensitivity trimmer	**5 Units**	**Consumable**
11	**High-Sensitivity Big Sound Module**	Large electret microphone capsule module with dual analog sound waveform and digital trigger outputs	**5 Units**	**Consumable**
12	**Capacitive Touch Module (TTP223B)**	Single-key capacitive digital touch switch; momentary / toggle output mode selectable; supply 2.0V–5.5V DC	**10 Units**	**Consumable**
13	**Touch Sensor Module (TTP22X)**	Multi-purpose capacitive contact sensor with low standby power; operating voltage 2.4V–5.5V DC	**6 Units**	**Consumable**
14	**Metal Touch Sensor Module (KY-036)**	Metal touch sensor with Darlington transistor and comparator; senses human skin touch on metal lead	**6 Units**	**Consumable**
15	**12-Channel Capacitive Touch (MPR121)**	MPR121 capacitive touch controller; 12 independent sensing inputs; I2C digital interface; 3.3V logic	**5 Units**	**Consumable**
16	**Pulse Heart-Rate Sensor**	Optical photoplethysmography (PPG) pulse sensor with built-in ambient noise cancellation; 3.3V–5V DC	**5 Units**	**Consumable**
17	**Color Recognition Sensor (TCS3200)**	TCS3200 programmable color light-to-frequency converter with 4 white LEDs; reads Red, Green, Blue, Clear filters	**5 Units**	**Consumable**
18	**ADXL345 3-Axis Digital Accelerometer Module**	3-axis digital accelerometer; high resolution (13-bit) measurement up to ± 16g; ultra-low power; SPI and I2C digital interfaces; 3.3V/5V compatible	**5 Units**	**Consumable**
19	**MPU6050 6-Axis Gyroscope and Accelerometer Sensor Module with I2C Interface and Digital Processor, for Robotics, Metal and Plastic**	6-Axis motion tracking module integrating 3-axis gyroscope (± 250, ± 500, ± 1000, ± 2000°/s) and 3-axis accelerometer (± 2g, ± 4g, ± 8g, ± 16g) with onboard Digital Motion Processor (DMP); I2C interface; operating voltage 3.3V–5V DC with onboard low-dropout regulator; essential for self-balancing robots, gesture navigation, flight stabilization, metal and plastic chassis robotics	**10 Units**	**Consumable**
20	**3-Axis Digital Magnetometer**	3-axis magnetoresistive compass module (HMC5883L / QMC5883L); I2C interface; 12-bit ADC; ± 8 Gauss range	**10 Units**	**Consumable**
21	**Laser Diode Emitter Module**	650nm red point laser diode module with onboard current-limiting resistor; output 5mW; 5V DC	**10 Units**	**Consumable**
22	**MQ-2 Smoke & Combustible Gas Sensor**	Wide-spectrum gas sensor module for methane, LPG, smoke, hydrogen; 5V heater with analog/digital comparator outputs	**3 Units**	**Consumable**
23	**MQ-3 Alcohol Vapor Gas Sensor**	High sensitivity to ethanol alcohol vapor; ideal for breathalyzer circuits; onboard LM393 comparator	**3 Units**	**Consumable**

Item #	Sensor Module Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
24	**MQ-4 Methane & Natural Gas Sensor**	High sensitivity to methane (CH_4) and natural gas leaks; fast response; onboard LM393 comparator	**3 Units**	**Consumable**
25	**MQ-5 LPG & Town Gas Sensor**	High sensitivity to LPG and compressed natural gas (CNG); onboard LM393 comparator	**3 Units**	**Consumable**
26	**MQ-6 Propane & Butane Gas Sensor**	Specialized high sensitivity for domestic LPG and butane gas cylinder leak detection; onboard LM393 comparator	**3 Units**	**Consumable**
27	**MQ-7 Carbon Monoxide (CO) Sensor**	High sensitivity to toxic carbon monoxide gas; dual-temperature cycling heater; onboard LM393 comparator	**3 Units**	**Consumable**
28	**MQ-8 Hydrogen Gas Sensor**	Specialized sensitivity to pure hydrogen gas (H_2); low sensitivity to alcohol/smoke; onboard LM393 comparator	**3 Units**	**Consumable**
29	**MQ-135 Air Quality & Pollution Sensor**	Environmental air quality detector for ammonia (NH_3), NO_x, alcohol, benzene, and smoke; onboard LM393	**3 Units**	**Consumable**

04-DISPLAYS-AND-INPUT-MODULES

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**0.96-inch I2C OLED Display Module (SSD1306)**	128×64 monochrome graphic OLED display; SSD1306 driver; 4-pin interface (VCC, GND, SCL, SDA); 3.3V–5V DC; high-contrast text and graphics *(Directly connects to Arduino, ESP32, and Raspberry Pi)*	**15 Units**	**Consumable**
02	**Robocraze 0.91 inch Blue OLED Display Module**	128×32 ultra-compact blue OLED display; I2C/IIC serial interface (VCC, GND, SCL, SDA); 3.3V–5V DC compatible; SSD1306 driver; for Arduino, ESP32, and Raspberry Pi	**10 Units**	**Consumable**
03	**4-Digit 7-Segment Display Module (TM1637)**	0.36-inch 4-digit red display with colon dots; onboard TM1637 driver; 2-wire serial bus (CLK, DIO); 5V/3.3V	**6 Units**	**Consumable**
04	**8×8 LED Dot Matrix Module (MAX7219)**	64 bright red LEDs in 8×8 matrix; MAX7219 serial driver; 5-pin SPI interface; daisy-chainable	**5 Units**	**Consumable**
05	**LDR Light Sensor Module**	5mm photoresistor with onboard LM393 voltage comparator, sensitivity trimmer pot, and dual Digital/Analog outputs (3.3V–5V compatible)	**10 Units**	**Consumable**
06	**Voice Recording & Playback Module (ISD1820)**	ISD1820 chip board; 10-second voice recording; onboard microphone; playback pushbuttons; includes 8Ω 0.5W speaker	**4 Units**	**Consumable**
07	**Dual-Axis Analog Joystick Module**	Dual 10 kΩ potentiometers for X and Y axes with integrated momentary pushbutton switch; spring auto-return to center	**5 Units**	**Consumable**
08	**4× 4 Matrix Membrane Keypad**	16-key matrix keypad (keys 0–9, *, #, A–D); flexible flat ribbon cable with 8-pin 2.54mm pitch female socket	**10 Units**	**Consumable**

05-MOTORS-SERVOS-DRIVERS-AND-RELAYS

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**TB6612FNG Dual Motor Driver Module for Arduino/Other Microcontroller**	**High-Efficiency Dual MOSFET H-Bridge Driver Module**; 1.2A continuous / 3.2A peak per channel; ultra-low internal resistance; runs cool without heatsink; operating logic 2.7V–5.5V; motor voltage 4.5V–13.5V; PWM speed control *(Replaces outdated L293D / L298)*	**10 Units**	**Consumable**
02	**SP Electron L293D Motor Driver Shield For Microcontrollers**	4-Channel driver shield using bipolar pulse-width modulation; plugs directly onto Arduino Uno / Mega headers; drives up to 4 DC motors or 2 stepper motors with screw terminal power connections	**5 Units**	**Consumable**
03	**L293D Dual H-Bridge Motor Driver Module**	Standalone BJT-based dual H-bridge motor driver board; controls speed and direction of 2 DC motors or 1 stepper motor; up to 600mA per channel; operating voltage 4.5V–12V; legacy demonstration driver	**2 Units**	**Consumable**
04	**L298P High-Power Motor Shield**	Arduino Uno form-factor shield featuring L298 dual full-bridge BJT driver; drives up to 2 DC motors (2A per channel); heavy screw terminals for external motor power; onboard buzzer and servo header interface	**2 Units**	**Consumable**
05	**PCA9685 16-Channel 12-bit PWM / Servo Driver**	16-channel independent PWM controller with I2C bus; driven by only 2 microcontroller pins (SDA, SCL); 12-bit resolution; handles 16 servos	**3 Units**	**Consumable**
06	**ULN2003 Stepper Motor Driver Board with 28BYJ-48 Stepper**	5V unipolar 4-phase stepper motor paired with ULN2003 Darlington driver board with 4-phase status indicator LEDs	**10 Sets**	**Consumable**
07	**1-Channel 5V/3.3V Relay Module with Optocoupler & Trigger Selector**	5V DC relay module with PC817 optocoupler isolation, transistor driver, status LED, and **High/Low level trigger selection jumper**; 10A 250V AC	**6 Units**	**Consumable**
08	**2-Channel 5V/3.3V Relay Module with Optocoupler & Trigger Selector**	Dual 5V DC relay board with independent optoisolated inputs, status LEDs, High/Low trigger selection jumpers, and NO/COM/NC terminals	**15 Units**	**Consumable**
09	**4 Channel 5V Relay Module with Optocoupler Isolation**	4-Channel 5V relay module with optocoupler isolation (10A 250V AC / 30V DC), Songle SRD-05VDC-SL-C with individual status LEDs for UNO, Raspberry Pi, ESP32, robotics & home automation	**5 Units**	**Consumable**
10	**Robocraze 8 Channel Relay Module 5V with Optocoupler Isolation**	Arduino Compatible PLC Control Board; 8 independent 10A high-current relays with optocoupler isolation, status indicator LEDs, and screw terminal blocks for smart home automation, IoT, robotics & industrial control	**3 Units**	**Consumable**
11	**150 RPM Plastic BO Gear Motor with Wheel**	3V–9V DC dual-shaft straight gearbox motor (150 RPM output speed @ 6V) complete with matching 65mm rubber-tread wheel	**30 Pairs**	**Consumable**
12	**150 RPM BO Motor (Standalone)**	3V–9V DC battery-operated single-shaft plastic gear motor (150 RPM)	**10 Units**	**Consumable**
13	**12V High-Torque DC Geared Motor**	12V DC heavy-duty metal-gear motor; 150 RPM output; 6mm D-shaped center output shaft	**10 Units**	**Consumable**
14	**Submersible Mini Water Pump**	12V DC, 3W submersible centrifugal brushless water pump with outlet nozzle; max head 1.5 metres	**3 Units**	**Consumable**
15	**Micro Positional Servo — Plastic Gear (SG90)**	9g micro positional servo motor; 0° - 180° pulse angle rotation; operating voltage 4.8V - 6.0V; stall torque 1.8 kg-cm; horns included	**30 Units**	**Consumable**
16	**Metal-Gear Micro Servo (MG90S)**	High-durability micro servo with brass and aluminum gears; 0° - 180° rotation; stall torque 2.2 kg-cm	**10 Units**	**Consumable**
17	**Standard High-Torque Positional Servo**	Full-size metal-gear positional servo (MG995 / MG996R); torque 10–12 kg-cm; double ball bearings; 0° - 180° control	**5 Units**	**Consumable**
18	**Continuous Rotation Metal Servo (360˚)**	Full-size continuous rotation servo (360° speed/direction control); high-torque metal gear train	**5 Units**	**Consumable**
19	**3-Channel Servo & ESC Tester (CCPM)**	Standalone digital servo tester with 3 modes (Manual speed dial, Neutral auto-center, Automatic window sweep); tests up to 3 servos simultaneously	**3 Units**	**Consumable**

06-WIRELESS-AND-IOT-MODULES

Item #	Module Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**NRF24L01+PA+LNA Wireless RF Transceiver Module Long Range 1100M with SMA Antenna (2-Pack)**	Long-range 2.4 GHz digital packet radio transceiver with Power Amplifier (PA) + Low Noise Amplifier (LNA) and **external SMA duck antenna**; 2 Mbps low power signal amplifier; up to 1,100 metres line-of-sight range; SPI bus; for Uno, Arduino & ESP32 wireless communication projects	**5 Packs (10 Units)**	**Consumable**
02	**Bluetooth Low Energy (BLE) Module (HM-10 / AT-09)**	2.4 GHz Bluetooth 4.0/5.0 BLE transceiver; UART serial; compatible with **both Android and Apple iOS smartphones**; 3.3V/5V tolerant	**10 Units**	**Consumable**
03	**Bluetooth Serial Module (HC-05)**	2.4 GHz Bluetooth V2.0+EDR classic serial module; SPP serial profile; UART AT command set; configurable Master/Slave mode; 3.3V logic (5V VCC tolerant); legacy Android/PC serial telemetry	**2 Units**	**Consumable**
04	**433 MHz / 315 MHz RF Transmitter & Receiver Pair**	ASK wireless super-regenerative RF transmitter and receiver pair; 315/433 MHz ISM band; operating voltage 3.3V–5V DC; range 20–100m for basic wireless telemetry and remote control experiments	**2 Pairs**	**Consumable**
05	**GSM / GPRS Cellular Module (SIM900)**	Quad-Band 850/900/1800/1900 MHz cellular shield; onboard standard SIM socket; UART serial AT command control; external antenna; 2G cellular SMS and telemetry demonstration	**2 Units**	**Consumable**
06	**RFID Reader Module & Tag Kit (RC522)**	13.56 MHz contactless RFID/NFC reader board (MFRC522); SPI bus; includes 1x RFID contactless card and 1x key fob tag	**10 Kits**	**Consumable**
07	**GPS Satellite Module (u-blox GY-NEO6MV2 / NEO-8M)**	High-sensitivity satellite receiver module with active ceramic patch antenna; EEPROM config retention; 9600 baud serial NMEA output	**3 Units**	**Consumable**

07-BREADBOARDS-WIRES-BATTERIES-AND-POWER

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**TP4056 USB-C 5V 1A Lithium Battery Charging Module**	Lithium battery charging board with **dual protection IC** (DW01A + 8205A MOSFET); 18650 Li-ion / LiPo charger with overcharge, overdischarge & overcurrent protection; USB-C input	**8 Units**	**Consumable**
02	**ROBOWAY 4S 4A Polymer Lithium Ion Charger Type-C (Pack of 4)**	14.8V 18650 battery charging module; USB-C input with boost converter; LED charging status indicator; integrated overcharge and balanced protection for 4-series Li-ion battery packs	**2 Packs (8 Units)**	**Consumable**
03	**Electronic Spices LM2596 DC-DC Step Down Power Supply Module (3A, Pack of 2)**	High-efficiency 3A switching voltage regulator buck converter; wide input voltage 4.0V–40V DC; adjustable output 1.3V–37V DC (supports 24V, 12V, 5V, 3.3V/3V rails); high efficiency (>90%); multi-turn trimmer pot; powers ESP32, Arduino, servos & motors from batteries	**5 Packs (10 Units)**	**Consumable**
04	**Robocraze SD Card Module for Arduino SPI Interface (3.3V/5V Level Shifter)**	SPI Interface Micro SD / MicroSDHC Reader Writer Module with **onboard 3.3V/5V Level Shifter**; data logging and storage expansion board for ESP32, Arduino, and Raspberry Pi robotics & IoT projects	**5 Units**	**Consumable**
05	****Robocraze DS3231 RTC Module **	AT24C32 Real Time Clock Memory Board**	**ESP32 / Arduino / Raspberry Pi**	**5 Units**
06	**Half-Size Solderless Breadboard (400 Tie-Points)**	400 tie-point solderless breadboard with self-adhesive backing tape; dual power distribution buses	**30 Units**	**Consumable**
07	**Full-Size Solderless Breadboard (800 Tie-Points)**	830 tie-point solderless breadboard; dual center terminal strip with 4 power distribution rails; side interlocking tabs	**20 Units**	**Consumable**
08	**Self-Adhesive Mini Proto Shield Breadboard**	Compact 170-point mini breadboard designed for Arduino Proto-Shield mounting	**15 Units**	**Consumable**
09	**General-Purpose Perfboard (Zero-PCB) — Medium**	FR2 single-sided copper clad dotted prototype board; size 80× 55 mm; 2.54mm hole spacing	**30 Units**	**Consumable**
10	**General-Purpose Perfboard (Zero-PCB) — Large**	FR2 single-sided copper clad dotted prototype board; size 140× 90 mm; 2.54mm hole spacing	**20 Units**	**Consumable**
11	**Male Breakable Pin Headers (Berg Strips)**	40-pin single row male headers; standard 2.54mm pitch; gold/tin plated; snappable to any pin length	**30 Strips**	**Consumable**
12	**Female Pin Headers (Berg Strips)**	40-pin single row female socket headers; standard 2.54mm pitch; straight mount	**30 Strips**	**Consumable**
13	**Serplex 560pcs Jumper Wire Kit (14 Lengths, 2–125mm)**	High-quality pre-formed U-shape solderless breadboard jumper wire kit; 14 different standard wire lengths (2mm to 125mm); flexible insulated copper core in segmented plastic organizer box	**5 Kits**	**Consumable**
14	**Jumper Wires — Male to Male (M-M)**	20cm flexible multicolored 40-pin ribbon jumper wires with molded square male pins	**5 Packs (200 Wires)**	**Consumable**
15	**Jumper Wires — Male to Female (M-F)**	20cm flexible multicolored 40-pin ribbon jumper wires with male pin to female socket	**5 Packs (200 Wires)**	**Consumable**
16	**Jumper Wires — Female to Female (F-F)**	20cm flexible multicolored 40-pin ribbon jumper wires with female socket to female socket	**3 Packs (120 Wires)**	**Consumable**
17	**Alligator Clip Test Leads**	Double-ended insulated alligator clip jumper cables; 12-inch length; multicolored jackets	**3 Packs (30 Leads)**	**Consumable**
18	**Hook-Up Wire Spools**	Single-core insulated hook-up wire spools (100 metres Red + 100 metres Black per set)	**2 Sets (4 Spools)**	**Consumable**
19	**Electronic Spices 2 x 3.7V 18650 Battery Holder with Wire Leads**	Dual 18650 Li-ion battery holder case (series connection for 7.4V DC nominal output) with color-coded positive/negative wire leads	**15 Units**	**Consumable**
20	**Electronic Spices 18650 Battery Holder 3-Slot with Wire Leads**	3-slot 18650 lithium battery holder box (3.7V × 3 = 11.1V / 12V output) with heavy-duty wire leads for robotics and motor supply	**10 Units**	**Consumable**

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
21	**Electronic Spices Li-ion Battery Universal Charger (Dual 3.7V 18650)**	Universal dual-slot desktop battery charger for 3.7V 18650 cylindrical lithium rechargeable cells with automatic cut-off and LED status	**3 Units**	**Consumable**
22	**AC to DC 4-Slot 18650 Battery Charger with LED Indicator**	Heavy-duty 4-slot intelligent AC mains charger for 3.7V rechargeable Li-ion 18650 batteries; independent channel charging with LED indicators	**2 Units**	**Consumable**
23	**HD-8993A 8-Slot Battery Charger with LCD Display**	8-slot multi-chemistry intelligent battery charger with LCD status display; 6400mA max output across independent channels; supports 18650, 26650, AA, AAA, Ni-MH, and 3.7V Lithium-ion cylindrical rechargeable batteries	**1 Unit**	**Consumable**
24	**3.7V 1200mAh LiPo (Lithium Polymer) Rechargeable Battery**	3.7V 1S flat pouch rechargeable lithium polymer cell with built-in Protection Circuit Module (PCM) preventing overcharge/overdischarge; pre-soldered leads / 2-pin JST connector; ideal for wearable electronics, IoT sensors, Seeed Studio XIAO ESP32-S3 battery charging, and TP4056 single-cell charging circuits	**30 Pieces**	**Consumable**
25	**18650 3.7V Li-ion (Lithium-Ion) Rechargeable Battery Cell**	Standard industrial 18650 cylindrical rechargeable lithium-ion cells (18× 65 mm); high-capacity (2000-2600 mAh) flat-top / button-top; fits 2-slot (7.4V) and 3-slot (11.1V) battery holders, desktop 4-slot chargers, robotics chassis, and high-drain BO motors	**30 Pieces**	**Consumable**
26	**9V Battery Snaps with DC Barrel Plug**	Heavy-duty molded hard-plastic 9V battery connector clips terminating in a 5.5mm/2.1mm DC barrel jack	**20 Units**	**Consumable**
27	**USB Programming Cables — Type-A to Type-B**	High-shielded 1.2m USB 2.0 cable for Arduino Uno and Arduino Mega	**10 Cables**	**Consumable**
28	**USB Programming Cables — Type-A to Mini-B**	High-shielded 1.0m USB 2.0 cable for Arduino Nano and microcontroller boards	**10 Cables**	**Consumable**
29	**USB to DC Barrel Jack Converter Cables**	5V USB Type-A to 5.5mm × 2.1mm center-positive DC plug cable for running breadboards from power banks	**5 Cables**	**Consumable**
30	**High-Capacity Portable Power Banks**	10,000 mAh dual-port (5V 2.1A / 1A) portable rechargeable power bank with battery LED meter	**3 Units**	**Consumable**
31	**Regulated DC Wall Power Adapter — 5V 2A**	230V AC to 5V DC, 2000mA regulated switch-mode wall power adapter with 5.5mm DC barrel jack	**5 Units**	**Consumable**
32	**Regulated DC Wall Power Adapter — 12V 2A**	230V AC to 12V DC, 2000mA regulated switch-mode wall power adapter with 5.5mm DC barrel jack	**5 Units**	**Consumable**
33	**Workshop Power Extension Boards**	Heavy-duty 5-socket universal surge-protected power extension strip with master switch, indicator, and 5m cord	**3 Units**	**Consumable**
34	**Twisted-Pair Mains Distribution Wires**	Red-Black twisted flexible copper electrical distribution wire	**20 Metres**	**Consumable**
35	**AC/DC Lamp Bulb Holders**	Standard insulated screw/bayonet socket bulb holders for bench load experiments	**5 Units**	**Consumable**
36	**FEDUS 12V 1.5Amp DC Power Supply Adapter Converter**	12 Volt 1.5A SMPS regulated power adapter converter (100–240V AC input, 12V 1.5A DC output with standard 5.5mm DC barrel jack); for CCTV cameras, routers, DVR/NVR, LCD monitors, LED strips, motors and microcontrollers	**2 Units**	**Consumable**

08-WORKSHOP-TOOLS-AND-MACHINES

Item #	Equipment / Machine / Tool Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**Mini Mic Pro (Latest Model – #1 Microphone for iPhone & Android, Wireless Mini Microphone, Clear Voice, Noise Cancelling, Lavalier Mic for Coding assistant)**	Wireless mini lavalier microphone system for iPhone & Android; active noise cancelling, ultra-low latency, clear voice pickup; optimized for AI coding assistant and workshop demos	**2 Units**	**Permanent Asset**
02	**Digital Multimeter (DMM)**	7 functions, 19 ranges; DC voltage (200mV - 1000V), AC voltage (200V - 750V), DC current (200 µA - 10A), resistance (200 Ω - 2 MΩ), diode test, continuity buzzer	**3 Units**	**Permanent Asset**
03	**Serplex Pen Type Digital Multimeter (4000 Count AC/DC 600V)**	Slim pen-style auto-ranging digital multimeter; 4000 counts; measures AC/DC voltage up to 600V, resistance, continuity buzzer, frequency, NCV non-contact voltage sensor with flashlight	**2 Units**	**Permanent Asset**
04	**Continuity Tester with Light Multi-Purpose Tool with Green LED**	Heavy-duty diagnostic continuity test tool with bright green LED indicator and tone buzzer; supplied complete with 2 AAA batteries; for wire tracing and PCB fault diagnosis	**1 Unit**	**Permanent Asset**
05	**Digital Voltage Tester Pen**	Contactless and contact AC voltage detector pen with audible alarm and bright LED indicator	**1 Unit**	**Permanent Asset**
06	**Digital Vernier Caliper**	150mm / 6-inch electronic digital vernier caliper; stainless steel frame; resolution 0.01mm / 0.0005"; zero-calibration key	**2 Units**	**Permanent Asset**
07	**Stainless Steel Workshop Ruler**	150mm / 12-inch heavy-duty stainless steel measuring rule with photo-etched metric and imperial scales	**3 Units**	**Permanent Asset**
08	**Auto-Retracting Measuring Tape**	5m × 19mm steel measuring tape with rubberized shockproof armor, belt clip, and thumb lock	**2 Units**	**Permanent Asset**
09	**Workshop Spirit Level**	12-inch aluminum magnetic spirit level with 3 shockproof acrylic vials (horizontal, vertical, 45°)	**1 Unit**	**Permanent Asset**
10	**Temperature-Controlled Soldering Stations**	15W–30W / 230V variable temperature soldering stations (280°C - 450°C) with heavy cast-iron cradle stand	**2 Stations**	**Permanent Asset**
11	**Helping Hand Magnifier Soldering Stand with 2 LED Lights**	PCB soldering auxiliary stand with dual adjustable alligator clips, heavy cast-iron base, flexible joints, bright dual LED illuminator, and dual magnification optical lens (35X / 12X, 65mm / 17mm)	**3 Units**	**Permanent Asset**
12	**Circuit Board Welding Repairing Disassembly Fixture (Motherboard Holder)**	Precision adjustable PCB and motherboard holding clamp with 360° rotation fixture, high-temperature resistance, and dual movable slide-locks for soldering, desoldering, and component rework	**2 Units**	**Permanent Asset**
13	**SP Electron Anti-Static Heat Insulation Silicone Magnetic Mat**	Large heat-resistant (500°C) silicone work mat for soldering and electronics repair; built-in screw sorting notches, tool holders, and magnetic zones for mobile, PCB & laptop assembly	**3 Units**	**Permanent Asset**
14	**Metal Wire Welding Soldering Tip Cleaner (Steel Ball Mesh)**	Heavy-duty copper/steel ball wire mesh tip cleaner with domed metal holder; cleans solder tips without lowering tip temperature or thermal shock	**2 Units**	**Permanent Asset**
15	**High-Vacuum Desoldering Pump**	Aluminum body spring-loaded manual solder sucker with high-suction Teflon nozzle	**5 Units**	**Permanent Asset**
16	**Lifelong 105 Pcs Electric Cordless Screwdriver Kit**	Precision mini electric screwdriver with forward-reverse function, adjustable torque, 1600mAh rechargeable Li-ion battery, USB-C charging cable, and 105 CR-V precision bit set for electronics, mobile & laptop repair	**1 Unit**	**Permanent Asset**
17	**Industrial Hot Melt Glue Guns**	High-temperature electric glue gun (230°C - 280°C) for 11mm standard glue sticks; fold-out wire stand and insulated nozzle	**2 Guns**	**Permanent Asset**
18	**Dual-Temperature Electric Heat Gun**	1800W heavy-duty industrial heat gun with dual-heat rocker switch (300°C / 500°C) and concentration nozzle attachment	**1 Unit**	**Permanent Asset**
19	**Adjustable Wire Stripper & Cutter**	Drop-forged steel wire stripper and diagonal cutter with adjustable brass gauge stop screw	**2 Units**	**Permanent Asset**
20	**Workshop Mini Plier Set (6-Piece)**	4.5-inch carbon steel precision pliers set: Long-nose, bent-nose, needle-nose, combination mini plier, external circlip plier, and wire-stripper	**1 Set**	**Permanent Asset**

Item #	Equipment / Machine / Tool Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
21	**Anti-Static Precision Tweezer Set**	6-piece non-magnetic anti-static ESD-safe stainless steel tweezers in zippered pouch (straight, curved, spade tips)	**1 Set**	**Permanent Asset**
22	**Jeweler Precision Screwdriver Set**	6-piece precision screwdriver set: Phillips PH0, PH1, and Flathead 1.4mm, 2mm, 2.4mm, 3mm in indexed case	**1 Set**	**Permanent Asset**
23	**40-Piece Multi-Bit Screwdriver Set**	Ratcheting screwdriver handle with flexible extension shaft and 40 magnetic bits (Torx, Hex, Phillips, Slotted, Pozidriv)	**2 Sets**	**Permanent Asset**
24	**Combination Spanner Set (12-Piece)**	Chrome vanadium drop-forged open and ring combination spanners (metric sizes 4mm to 30mm) with wall-mountable rack	**1 Set**	**Permanent Asset**
25	**Universal Adjustable Wrench**	8-inch drop-forged carbon steel adjustable wrench with laser-etched metric jaw opening scale	**1 Unit**	**Permanent Asset**
26	**Imperial Allen Hex Key Set**	10-piece ring imperial Allen hex key set for machine hex socket screws	**1 Set**	**Permanent Asset**
27	**Table-Top Bench Vice**	Heavy-duty cast-iron bench vice with serrated steel jaws, swivel base, locking lever, and integrated anvil surface	**1 Unit**	**Permanent Asset**
28	**Metal Screw C-Clamps**	3-inch heavy-duty forged steel screw-threaded C-clamps with swivel shoe	**2 Units**	**Permanent Asset**
29	**12-inch Heavy-Duty Hacksaw Frame**	Tubular steel hacksaw frame with cast handle and wingnut blade tensioner for 12-inch blades	**1 Frame**	**Permanent Asset**
30	**6-inch Mini Hacksaw Frame**	Compact aluminum alloy mini hacksaw frame for tight spaces and delicate cutting	**1 Frame**	**Permanent Asset**
31	**Engineering Hand File Set (6-inch)**	6-piece high-carbon steel file set: flat, hand, round, half-round, triangular, and square with non-slip rubber handles	**1 Set**	**Permanent Asset**
32	**Diamond Needle File Set (3-inch)**	6-piece miniature diamond needle file set for fine deburring of plastic boxes, project enclosures, and small chassis	**1 Set**	**Permanent Asset**
33	**Steel Shaft Claw Hammer**	0.75 kg polished carbon steel head with tubular steel shaft and vibration-absorbing rubber grip	**1 Unit**	**Permanent Asset**
34	**Engineer's Ball Pein Hammer**	0.75 kg forged carbon steel ball pein hammer for metal forming, tapping, and center punching	**1 Unit**	**Permanent Asset**
35	***BLACK+DECKER LD120VA-IN 20V MAX* POWERECONNECT Li-Ion Cordless Drill/Driver + 30 Piece Bits and Drivers, 1.5 Ah Battery & Charger Included, 750 RPM, 24-Position Clutch, Keyless Chuck, Soft-Grip Handle****	20V MAX cordless lithium-ion drill/driver; 750 RPM variable speed; 24-position clutch prevents stripping/overdriving screws; 3/8-inch keyless chuck; includes 20V 1.5 Ah POWERCONNECT battery, charger, and 30-piece accessory set with drill bits and screwdriver bits	**1 Unit**	**Permanent Asset**
36	**Bench Drill Press Stand & Holder**	Heavy cast-iron base drill press stand compatible with the 500W electric drill for perpendicular precision drilling	**1 Unit**	**Permanent Asset**
37	**HSS Twist Drill Bit Set (13-Piece)**	13-piece High-Speed Steel (HSS) drill bits (1 mm - 10 mm) in indexed steel storage index box	**1 Set**	**Permanent Asset**
38	**Cordless Rotary Multitool Set**	High-speed Li-ion rotary tool (up to 30,000 RPM) with flexible shaft, collets, cutting discs, grinding stones, and polishing wheels	**1 Set**	**Permanent Asset**
39	**Variable-Speed Electric Air Blower**	500W electric high-velocity air blower with rubber directional nozzle for bench cleaning and dust removal	**1 Unit**	**Permanent Asset**
40	**Flexible Self-Healing Cutting Mat**	A3 size (18× 12 inches) 5-ply self-healing cutting mat with printed metric grid lines, millimeters, and angle guides	**1 Unit**	**Permanent Asset**
41	**Modular Component Storage Cabinets**	High-impact plastic storage cabinets with 60 transparent drawers each (2× 2× 5 inches) for sorting small electronics	**2 Units (120 Drawers)**	**Permanent Asset**
42	**Wall-Mounted Pegboard Tool Systems**	3× 2 feet metal tool pegboards with modular hanging hooks, spanner racks, plier clips, and tool shelves	**2 Boards**	**Permanent Asset**
43	**BUILDSKILL 2 IN 1 Automatic Wire Stripping & Cutting Tool**	Precision 2-in-1 automatic wire stripper and cutter; adjustable pressure tension knob and wire length guide; ergonomic non-slip handle; for electrical, electronics, automotive cords and cables	**1 Unit**	**Permanent Asset**

Item #	Equipment / Machine / Tool Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
44	**Wire Crimping Tool Kit Ferrule Crimping Plier with 1200pcs Terminals**	Self-adjusting electrician ratchet ferrule crimper plier (AWG 23-7 / 0.25-10mm²) paired with 1200pcs copper crimp connector ferrule wire terminal assortment box	**1 Kit**	**Permanent Asset**
45	**Stainless Steel Heavy Quality Diagonal Nipper Wire Cutter**	Heavy-duty stainless steel diagonal nipper/flush wire cutter; precision cutting edges with spring return and insulated non-slip grips; for component lead clipping, electronics and jewelry wiring	**1 Pack (1 Unit)**	**Permanent Asset**
46	**VEGA Small General Cutting Scissor for Home, Office & Craft Use (SCS-01)**	Stainless steel multipurpose precision scissors with ergonomic comfort grip; for paper, cardboard, fabric, thread, tapes, and craft cutting	**1 Unit**	**Permanent Asset**
47	**Serplex® 8 Inch Heavy Duty Scissors Tin Snips with Stainless Steel Blade**	8-inch industrial heavy-duty scissors / tin snips with stainless steel hardened cutting blades; for cutting thin sheet metal, aluminum, tin, wire mesh, thick plastic, and cardboard	**1 Unit**	**Permanent Asset**
48	**Qatalitic Detail Pen Cutter with 5 Interchangeable Sharp Blades**	Precision craft detail pen knife / scalpel tool made of high-grade steel with textured anti-slip grip; supplied with 5 interchangeable surgical-sharp replacement blades; for fine model making, stencil trimming, paper craft, and PCB trace cutting	**2 Units**	**Permanent Asset**
49	**KABEER ART Heavy Duty 18 mm Cutter Set With 10 Replacement Blades**	Heavy-duty 18mm utility cutter knife with auto-lock slide mechanism and rugged body; packaged with 10 replacement high-carbon snap-off blades in storage case; for heavy craft, cardboard, acrylic, and foam boards	**1 Set**	**Permanent Asset**
50	**Verilux® Mini Cordless Rotary Tool Kit (3.7V, 5 Speeds, with 58 Accessories)**	3.7V rechargeable mini cordless rotary multitool with 5 variable speed levels; includes 58 accessories (collets, sanding bands, grinding stones, drill bits, polishing wheels) and USB charging cable; for delicate craft, PCB drilling, engraving, and polishing	**1 Kit**	**Permanent Asset**
51	**Hetkrishi Mini Die Grinder Rotary Tool Variable Speed (130W)**	Compact 130W electric mini die grinder with variable speed dial (8,000 - 30,000 RPM); heavy-duty motor for high-torque drilling, sanding, grinding, buffing, polishing, and engraving	**1 Unit**	**Permanent Asset**

09-WORKSHOP-CONSUMABLES-AND-MATERIALS

Item #	Consumable Item Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**Electronic Spices Quickfix Solder Paste Flux (Pack of 2, 50g+50g=100g)**	Premium brown rosin-rich active solder paste flux; facilitates fast solder wetting, prevents oxidation, and cleans copper traces during prototyping	**2 Packs (4 Tubs)**	**Consumable**
02	**Mechanic XG-50 Tin Solder Paste (Sn63/Pb37, 35g)**	Professional No-Clean soldering tin paste alloy (63\% Sn / 37\% Pb, melting point 183°C); ideal for SMD rework, BGA, PCB assembly & mobile repair	**2 Tubs**	**Consumable**
03	**SCHOFIC 5 Feet 3MM BGA Desoldering Wick**	High-absorption pure copper desoldering braid / solder remover wick (3.0mm width, 5 feet length); for cleaning solder bridges and SMD pads	**5 Spools**	**Consumable**
04	**Rosin-Core Solder Wire Spools**	100g spools of 60/40 leaded or lead-free 20/22 AWG rosin-flux-core soldering wire	**3 Spools (300g total)**	**Consumable**
05	**Hot Melt Glue Sticks**	Standard 11 mm× 200 mm high-clarity EVA hot melt adhesive glue sticks	**30 Sticks**	**Consumable**
06	**12-inch Hacksaw Replacement Blades**	12-inch high-carbon steel bi-metal flexible hacksaw blades (18/24 TPI)	**5 Blades**	**Consumable**
07	**6-inch Mini Hacksaw Replacement Blades**	6-inch high-carbon steel replacement mini blades (24 TPI)	**5 Blades**	**Consumable**
08	**Waterproof Sandpaper Assortment**	Silicon carbide waterproof sandpaper sheets: Grits 80, 100, 120, 200, 300	**3 Packs (15 Sheets)**	**Consumable**
09	**Machine Hardware Fasteners Kit**	Assorted metric machine screws, hex nuts, and washers: M4 × 12/20mm, M5 × 16/25mm, M6 × 20mm	**5 Sets**	**Consumable**
10	**Nylon Cable Ties (Zip Ties)**	Self-locking UV-resistant nylon cable ties: Small, Medium, Big (100 ties/pack)	**5 Packs (500 ties total)**	**Consumable**
11	**Heat Shrink Tubing Assortment**	Polyolefin flame-retardant heat shrink tubing assortment (diameters 2mm, 3mm, 5mm, 8mm, 10mm)	**3 Packs (60 Pieces)**	**Consumable**
12	**Prototyping Corrugated Cardboard Sheets**	Heavy-duty 3-ply corrugated cardboard sheets for rapid structural chassis mockups	**3 Sheets**	**Consumable**
13	**Prototyping Foam Core Boards**	Lightweight rigid 5mm white foam core boards for precision enclosure fabrication	**3 Boards**	**Consumable**
14	**Wooden Popsicle Craft Sticks**	Natural birch wood craft sticks for linkages, chassis beams, and structural assemblies	**1 Pack (100 Sticks)**	**Consumable**
15	**PVA Wood Adhesive Glue**	500 ml bottle of fast-drying non-toxic PVA wood glue	**1 Bottle**	**Consumable**
16	**Thick String & Heavy Rubber Bands**	20m spool of multicolored thick cord + 100 heavy-duty elastic rubber bands	**2 Spools + 1 Pack**	**Consumable**
17	**Polyolefin Heat Shrink Sleeve Wrapping Tube (1mm–10mm, 1m each)**	Polyolefin flame-retardant heat shrink tubing assortment in individual 1-meter lengths (diameters 1mm, 2mm, 3mm, 4mm, 5mm, 6mm, 7mm, 8mm, 9mm, 10mm; 2:1 shrink ratio); for DIY projects, wiring insulation, and cable repairs	**1 Set (10 Metres)**	**Consumable**
18	**Mario Masking Tape (12mm Width, 20m Length, Pack of 12 Rolls)**	12mm width × 20m length paper masking tape (pack of 12 rolls); easy-tear DIY craft tape, residue-free peeling; for labeling, carpentry, light holding, and painting	**1 Pack (12 Rolls)**	**Consumable**
19	**Pidilite Steelgrip Self-Adhesive PVC Electrical Insulation Tape**	Premium PVC electrical insulation tape (6m length per roll); flame-retardant, high-adhesion dielectric insulation for wire splicing and electrical joints; pack of 12 rolls	**1 Pack (12 Rolls)**	**Consumable**
20	**Instamall Aluminum Gas Lighter Refill Can (100 ML, Pack of 2)**	Refined liquefied butane gas refill canisters (100 ml each, pack of 2 red aluminum cans); for refilling micro gas torches, flame tools, bench lighters, and heat shrink activation	**1 Pack (2 Cans)**	**Consumable**

10-SAFETY-EQUIPMENT-AND-FIRST-AID

Item #	Safety Item Name	Technical Specifications & Safety Ratings	Required Quantity	Asset Classification
01	**Clear Safety Goggles**	Polycarbonate high-impact scratch-resistant safety goggles (EN166 / ANSI Z87.1 certified); un-tinted clear lens	**20 Units**	**Permanent Asset**
02	**Protective Workshop Work Gloves**	Medium-size latex/nitrile coated work grip gloves for handling sharp sheet metals, drills, and hot tools	**5 Pairs**	**Consumable**
03	**Particulate Safety Dust & Fume Masks**	80% particulate filtration efficiency masks for soldering flux fumes and workshop wood/drill dust	**10 Units**	**Consumable**
04	**Comprehensive Lab First-Aid Kit**	Portable wall-mountable first-aid station with sterile bandages, burn gel dressing, antiseptic liquid, eyewash, and surgical tape	**1 Kit**	**Permanent Asset**
05	**Portable Fire Extinguishers (ABC Dry Powder)**	2 kg capacity ABC dry chemical powder portable fire extinguisher with pressure gauge; rated for electrical and material fires	**2 Units**	**Permanent Asset**